

FUSIONEVAL: BENCHMARKING MODEL FUSION FOR LARGE LANGUAGE MODELS

Shuo Cai^{1*}, Yanggan Gu^{1*}, Yuhang Liu^{1*}, Su Lu¹, Yuanyi Wang¹, Yang Yu¹,
Zhen Li¹, Zhitian Hou², Guanghao Zhu¹, Ming Li^{1,3†}, Hongxia Yang^{1,2,3†}

¹The Hong Kong Polytechnic University ²InfiX.ai

³PolyU-Daya Bay Technology and Innovation Research Institute

shuo1031.cai@connect.polyu.hk

ABSTRACT

Model fusion aims to integrate the capabilities of multi-expert models into a single model. Existing evaluations offer limited coverage of reasoning with long chains of thought and tool use, while strong average performance can mask substantial losses in individual expert capabilities. We introduce FUSIONEVAL, a benchmark that provides controlled expert pools, integrates fusion methods with and without additional training, and offers a unified evaluation suite. The pools contain domain experts independently trained from Qwen3 and Llama base models across math, code, science, instruction following, and tool use. The suite evaluates capability retention and computational cost on eight public benchmarks under comparable data access rules, reporting normalized average and worst performance, runtime, and peak GPU memory usage. Extensive experiments show that training-free methods preserve reasoning performance but substantially degrade tool use at both model scales. In contrast, MOPD preserves tool use and achieves a more balanced capability profile, but requires additional data preparation, inference, and training. Jointly retaining reasoning, instruction following, and tool use at low computational cost remains a challenge. We hope FUSIONEVAL supports future research on efficient model fusion that jointly preserves expert capabilities. Our code is available at https://github.com/Baicaihaochi/FusionEval_Benchmark.

1 INTRODUCTION

Model fusion integrates the knowledge and abilities of independently trained models into a single deployable model, reducing the need to deploy and maintain multiple source models (Wortsman et al., 2022; Yadav et al., 2023; Jin et al., 2023; Ma et al., 2026). The growing ecosystem of open large language models (LLMs) offers an increasingly diverse pool of reusable models, while fusion methods span weight combination, representation alignment, and multi-teacher distillation (Cai et al., 2026). Model fusion benchmarks provide a common setting for comparing fusion effectiveness and computational cost (Tang et al., 2025a; He et al., 2025).

Existing evaluations differ in their coverage of fusion methods, expert models, and capabilities, as summarized in Table 1. Merging at Scale examines the effects of model size, base model quality, and expert count, but focuses on fusion methods without additional training (Yadav et al., 2025). FusionBench supports methods both with and without additional training (Tang et al., 2025a), while compositional evaluations study skill combination through tasks such as question answering, summarization, and natural language inference (Tam et al., 2024). Beyond conventional NLP tasks, FusionBench and Model-GLUE include mathematical reasoning (Tang et al., 2025a; Zhao et al., 2024), and MergeBench standardizes expert training and evaluation across domains including math and code, reporting capability retention and runtime (He et al., 2025). Despite this progress, these evaluations provide limited coverage of experts trained on long chain-of-thought (CoT) responses and of tool use, leaving the joint retention of reasoning, instruction following, and tool use insuf-

*Equal contribution. †Corresponding authors.

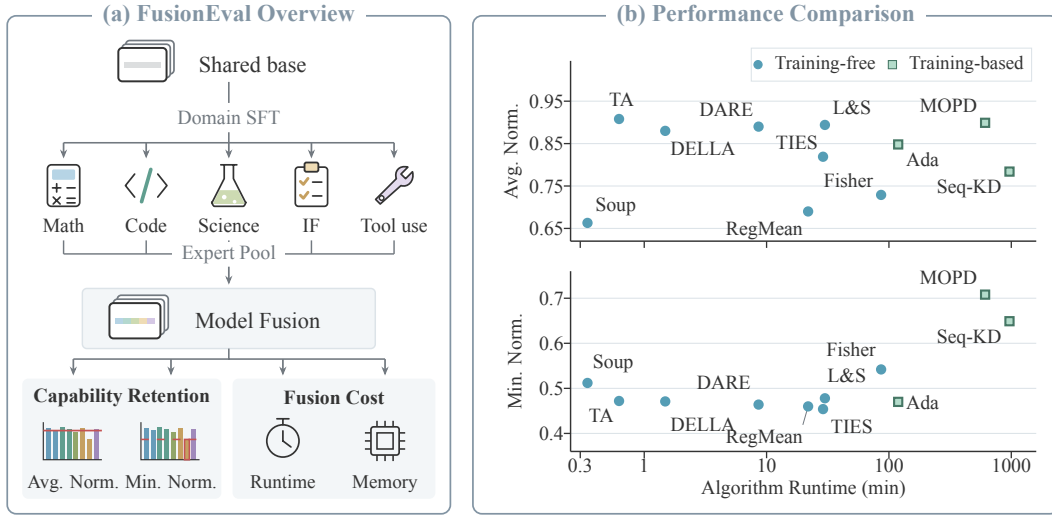


Figure 1: Overview of FUSIONEVAL. (a) Domain SFT of a shared base yields experts in five domains. Fusion is evaluated by capability retention, runtime, and peak GPU memory. (b) Qwen3-4B performance versus runtime (minutes, log scale). Avg. Norm. and Min. Norm. are the mean and minimum of benchmark scores normalized by the corresponding experts.

Table 1: Coverage of model fusion evaluations. Training-free and Training-based indicate support for fusion methods without and with gradient updates, respectively. Large model denotes evaluation of LLMs with at least 7B parameters. Reasoning covers math, code, or science tasks. Continual Fusion evaluates fusion as experts arrive sequentially. Tool use evaluates external tool calls.

Evaluation	Training-free	Training-based	Large model	Reasoning	Continual Fusion	Tool use
Compositional Eval (Tam et al., 2024)	✓	✓	✗	✗	✗	✗
Model-GLUE (Zhao et al., 2024)	✓	✗	✓	✓	✗	✗
FusionBench (Tang et al., 2025a)	✓	✓	✓	✓	✓	✗
Merging at Scale (Yadav et al., 2025)	✓	✗	✓	✗	✗	✗
MergeBench (He et al., 2025)	✓	✓	✓	✓	✗	✗
FUSIONEVAL	✓	✓	✓	✓	✓	✓

ficiently tested. A complementary evaluation setting is therefore needed to examine whether these capabilities can be preserved simultaneously and at what computational cost.

To address these limitations, we introduce FUSIONEVAL, a benchmark that provides controlled expert pools, integrates fusion methods with and without additional training, and offers a unified evaluation suite for capability retention and computational cost. As shown in Figure 1, we use open models from the Qwen (Yang et al., 2025a) and Llama (Meta AI, 2024) families as bases and select five domains: math, code, science reasoning, instruction following, and tool use. We select datasets for each domain (Table 2) and independently apply supervised fine-tuning (SFT) to each base model using the corresponding data, producing a separate pool of five domain experts for each base checkpoint. Unlike M2RL, which obtains domain experts through reinforcement learning from a shared model trained with mixed SFT data (Wang et al., 2026), we train each expert directly from the base model through SFT. This setting reflects the practical scenario of training experts separately for different domains and tests whether fusion retains each expert’s capabilities. We compare fusion methods using the same expert pools, consistent data access rules, and eight public benchmarks. The evaluation suite normalizes each benchmark score by that of the corresponding domain expert and reports the mean and minimum across benchmarks to assess overall retention and the weakest retained performance. It also reports fusion runtime and peak GPU memory.

Our experiments show that several fusion methods retain much of the math and code performance of experts trained on long CoT responses, but retention is uneven across reasoning, instruction following, and tool use: high average scores can mask severe losses in tool use. In contrast, multi-teacher

on-policy distillation (MOPD) achieves the highest minimum normalized benchmark score among the evaluated methods at both tested Qwen3 scales, yielding a more balanced capability profile. However, it requires additional data preparation, inference, and training. We hope FUSIONEVAL encourages the development of efficient model fusion methods that jointly preserve expert capabilities.

2 RELATED WORK

Model Fusion Methods. Model fusion combines the capabilities of multiple models through parameter operations or distillation of model outputs. Training-free fusion starts with weight averaging and task-vector addition, as in Model Soup (Soup) and Task Arithmetic (TA) (Wortsman et al., 2022; Ilharco et al., 2023). To reduce interference between expert updates, TIES resolves sign conflicts, while DARE, DELLA, and Dataless Localize-and-Stitch (L&S) control which updates are combined through random dropping or magnitude-based selection (Yadav et al., 2023; Yu et al., 2024; Deep et al., 2024; He et al., 2024). Other training-free methods use data statistics to guide parameter combination: Fisher Merging (Fisher) estimates parameter importance, RegMean and LOT-Merging (LOT) solve for weights using layer-input statistics, and FeatCal calibrates fused representations through closed-form updates (Matena & Raffel, 2022; Jin et al., 2023; Sun et al., 2025; Gu et al., 2026). Training-based fusion adjusts the result through optimization, either by learning fusion coefficients, as in AdaMerging (Ada), or by training representation calibration modules, as in Surgery (Yang et al., 2024b;a). Distillation directly optimizes a student model: Seq-KD learns from teacher-generated responses, GKD provides teacher supervision on student-generated sequences, and MOPD uses on-policy distillation to integrate multiple domain teachers (Kim & Rush, 2016; Agarwal et al., 2024; Ma et al., 2026).

Model Fusion Evaluations. Existing model fusion evaluations differ in how they compare methods and select experts. FusionBench supports comparisons across models and tasks through unified algorithm implementations, while Merging at Scale uses controlled experiments to study the effects of model size, base-model quality, and expert count (Tang et al., 2025a; Yadav et al., 2025). Expert sources further distinguish these settings: Model-GLUE studies the selection and combination of community models, whereas MergeBench trains domain experts under a common protocol to compare multi-task performance, pretrained knowledge retention, and runtime on consistent expert pools (Zhao et al., 2024; He et al., 2025). Task coverage has also expanded from conventional NLP to math and code, but these settings mainly use task- or instruction-tuned experts, leaving capability retention after fusing long-CoT experts insufficiently studied (Tang et al., 2025a; Zhao et al., 2024; He et al., 2025). Tool use also receives limited coverage in existing model fusion evaluations, leaving a need for systematic comparisons of capability retention when independently trained long-CoT and tool use experts are fused. FUSIONEVAL trains domain experts independently from a common base and compares fusion methods under consistent data-access conditions and evaluation tasks, reporting mean and worst-benchmark capability retention alongside computational cost.

3 FUSIONEVAL

FUSIONEVAL evaluates how well fused models retain domain-expert capabilities and how much computation fusion requires. We select tasks in reasoning, instruction following, and tool use (Section 3.1), and train domain experts independently from open-source base models (Section 3.2). Fusion methods use the same expert pool and are evaluated on eight public benchmarks under a common protocol (Section 3.3). We report mean and worst-benchmark normalized performance, runtime, and peak GPU memory (Section 3.4).

3.1 TASK SCOPE

FUSIONEVAL covers five task categories: math, code, science question answering, instruction following, and tool use. Math, code, and science question answering assess the reasoning capabilities of long-CoT experts; instruction following and tool use test compliance with explicit constraints and correct calls to external tools, respectively. We examine whether fused models that retain math,

Table 2: Datasets for SFT of domain experts. Sample counts are reported after preprocessing.

Domain	Training datasets	# Samples
Math	ODA-Math-460k; Superior-Reasoning-SFT (Math)	334.1K
Code	Nemotron-Competitive-Programming-v2	300K
Science	OpenScienceReasoning-2	249.9K
IF	Nemotron-Instruction-Following-Chat-v3	179.8K
Agent	Nemotron-Post-Training-Dataset-v1; HardGen; ToolMind; Nemotron-Agentive-v1	198.6K

code, and science reasoning capabilities can also follow instructions and call tools correctly, and which capabilities are more vulnerable to degradation during fusion.

3.2 EXPERT POOL CONSTRUCTION

Base models. We select advanced open-source models, Qwen3-4B-Base, Qwen3-8B-Base, and Llama-3.2-3B, as base models and construct an expert pool for each (Yang et al., 2025a; Meta AI, 2024). Each pool contains five experts trained independently by domain-specific SFT from the same base model, with the same architecture and tokenizer. Fixing the data and training settings for each domain allows controlled expert construction and comparable fusion results.

Training data. Table 2 summarizes the training datasets for each domain and the sample counts after preprocessing. We apply basic data cleaning, including format standardization, deduplication, and filtering of overlong samples ($> 32k$ tokens) and abnormally repetitive content. Before expert training, we remove training samples that exactly match evaluation samples in either raw or normalized text; Appendix Table 15 reports the counts. Each domain expert is trained separately by SFT on its corresponding domain data; detailed training settings are provided in Appendix Table 13. We use prompts from the datasets listed in Appendix Table 26 for distillation. In MOPD, the student generates responses and the corresponding domain expert provides token-level supervision (Ma et al., 2026); for Seq-KD, we train the student on responses generated in advance by domain experts (Kim & Rush, 2016).

3.3 EVALUATION PROTOCOL

We compare training-free and training-based fusion methods, including weight combination, data-assisted parameter fusion, and multi-teacher distillation. We also evaluate FeatCal and Surgery for post-fusion calibration. Table 3 lists how each method computes the fused model and what data it requires; notation is defined in Appendix B.2.

We evaluate domain experts and fused models on eight public benchmarks across five domains; Table 4 lists sample counts and generations per question. Models from the same base use the same prompts, decoding parameters, and scoring rules on each benchmark. Multi-turn tool tasks include issuing calls, receiving tool results, and continuing the interaction. We select hyperparameters and checkpoints through quick evaluation on five benchmarks, then evaluate the selected models on all eight benchmarks. Appendix B.4 details input templates, response extraction, and full settings.

3.4 METRICS

Benchmark scores. AIME24, AIME25, and GPQA-D use answer accuracy; LCBv5 and LCBv6 use pass@1 based on program tests (Jain et al., 2025; Rein et al., 2024); IFEval and IFBench use prompt-level strict accuracy (Zhou et al., 2023; Pyatkin et al., 2025). Repeated responses are scored individually and averaged, with generations per question listed in Table 4. BFCLv3 uses micro-averaged success. Raw Avg. is the equal-weight mean of the eight benchmark scores on the percentage scale.

Expert-normalized performance. We divide the score $s_b(\theta)$ of model θ on benchmark b by the corresponding domain-expert score $s_b^{\text{exp}} > 0$ from the same pool, and compute the following over

Table 3: Core operations and data requirements of evaluated fusion methods. Data access refers to model construction, excluding validation; inputs are unlabeled unless specified. RegMean is shown in its linear-layer form.

Method	Core operation	Data / signals
Training-free		
Soup	$\theta^* = M^{-1} \sum_i \theta_i$	Weights
TA	$\theta^* = \theta_0 + \lambda \sum_i \tau_i$	Weights
TIES	$\theta^* = \theta_0 + \lambda \text{TIES}_\rho(\{\tau_i\})$; trim/sign/mean	Weights
DARE	$\theta^* = \theta_0 + \lambda \sum_i \frac{(1-z_i) \odot \tau_i}{1-p}$; drop/rescale	Weights
DELLA	Magnitude dropout, rescaling, sign-aligned mean	Weights
L&S	$\theta_j^* = \theta_{0j} + \frac{\sum_i m_{ij} \tau_{ij}}{\max(1, \sum_i m_{ij})}$	Weights
Fisher	$\theta^* = \frac{\sum_i F_i \odot \theta_i}{\sum_i F_i + \epsilon}$	Inputs; Fisher estimates
RegMean	$W^* = (\sum_i \tilde{G}_i)^{-1} \sum_i \tilde{G}_i W_i$	Inputs; Gram matrices
FeatCal	Forward-order regularized closed-form weight updates	Inputs; expert features
Training-based		
Ada	$(\theta^*)^\ell = \theta_0^\ell + \sum_i \lambda_i^\ell \tau_i^\ell$; entropy minimization	Unlabeled inputs
Surgey	Train task-specific alignment modules	Inputs; expert features
Seq-KD	Sequence imitation of teacher responses	Prompts; responses
MOPD	Domain-teacher distillation on student rollouts	Prompts; token probabilities

Table 4: Evaluation benchmarks by domain. n denotes generations per question; slash-separated values follow the benchmark order.

Domain	Benchmarks	# Samples	n
Math	AIME24 / AIME25 (AIME, 2025)	30 / 30	16 / 16
Code	LCBv5 / LCBv6 (Jain et al., 2025)	279 / 131	2 / 4
Science	GPQA-D (Rein et al., 2024)	198	4
Instruction-following	IFEval / IFBench (Zhou et al., 2023; Pyatkin et al., 2025)	541 / 300	2 / 2
Agent	BFCLv3 (Patil et al., 2025)	4,441	1

$K = 8$ benchmarks (Ilharco et al., 2023; He et al., 2025):

$$\text{Avg. Norm.} = \frac{1}{K} \sum_{b=1}^K \frac{s_b(\theta)}{s_b^{\text{exp}}}, \quad \text{Min. Norm.} = \min_{1 \leq b \leq K} \frac{s_b(\theta)}{s_b^{\text{exp}}}. \quad (1)$$

Avg. Norm. measures mean performance relative to the experts, while Min. Norm. measures normalized performance on the worst benchmark to identify capability losses that the mean may hide. A normalized score of 1 matches the corresponding expert.

Computational cost. Following MergeBench (He et al., 2025), we measure Algorithm runtime and Peak GPU memory during fusion to assess computational cost. Appendix B.5 gives the measurement settings; Appendix B.6 reports search counts and cost estimates.

4 EXPERIMENTS

We first compare multi-domain performance on the Qwen3-4B and Qwen3-8B expert pools, focusing on mean performance and capability retention on the worst benchmark (Section 4.1). We then examine Continual Fusion (Section 4.2), and analyze fusion costs, parameter update patterns, and the effects of post-fusion calibration (Section 4.3).

4.1 MULTI-DOMAIN FUSION PERFORMANCE

We compare multi-domain fusion performance on the Qwen3-4B and Qwen3-8B expert pools (Tables 5 and 6). We also evaluate the Llama-3.2-3B expert pool to examine whether the findings hold in another model family; settings and results are provided in Appendix A.1. Data-assisted fusion uses 128 calibration examples per domain. We randomly sample 70% of each domain’s training data and combine them for SFT of the base model, producing Student as the distillation initialization and a comparison baseline. Distillation data and training configurations are provided in Appendices C and B.7. Non-distillation methods are ordered by Min. Norm., with distillation methods listed separately. Appendices B.6 and B.7 give search settings and method configurations; Appendix A reports cross-domain expert results, and Appendix B.9 gives full LOT and post-fusion calibration results.

Table 5: Model fusion performance on Qwen3-4B. Benchmark scores and Raw Avg. are percentages; Avg. Norm. and Min. Norm. are relative to the corresponding domain experts. Experts lists the per-benchmark reference scores, and Student is the mixed-domain SFT baseline. Bold marks the best fusion result in each column.

Method	Math		Code		Science	IF		Tool use	Raw	Norm.	
	AIME24	AIME25	LCBv5	LCBv6	GPQA-D	IFEval	IFBench	BFCLv3	Avg.	Avg.	Min.
Experts	73.4	67.4	44.4	41.6	51.9	86.0	38.6	66.0	58.7	1.000	1.000
TIES	69.0	64.4	41.6	35.9	36.9	65.3	36.2	30.0	47.4	0.819	0.454
RegMean	43.5	40.2	31.5	29.4	47.1	68.9	28.7	30.4	40.0	0.690	0.460
DARE	66.9	62.5	47.5	42.6	49.0	70.9	37.0	30.6	50.9	0.890	0.464
Ada	67.5	58.8	42.3	38.7	49.6	66.6	35.0	31.0	48.7	0.848	0.470
DELLA	68.5	64.8	47.0	40.8	49.5	64.4	36.0	31.1	50.3	0.880	0.471
TA	67.5	67.1	48.2	42.2	51.4	69.4	37.8	31.1	51.8	0.908	0.472
L&S	67.1	62.5	49.3	39.5	44.6	70.3	42.3	31.5	50.9	0.894	0.478
Soup	45.4	35.6	27.8	27.5	51.3	62.9	24.8	33.8	38.6	0.663	0.512
Fisher	39.8	38.1	38.2	31.9	47.1	63.9	29.6	45.1	41.7	0.729	0.542
Student	49.1	47.9	33.5	31.5	30.9	64.8	27.8	63.7	43.7	0.741	0.596
Seq-KD	47.9	43.8	36.4	31.9	34.5	67.7	37.5	63.7	45.4	0.784	0.649
MOPD	59.6	53.3	42.8	38.9	36.7	77.5	41.4	66.1	52.1	0.899	0.708

Table 6: Model fusion performance on Qwen3-8B. Benchmark scores and Raw Avg. are percentages; Avg. Norm. and Min. Norm. are relative to the corresponding domain experts. Experts lists the per-benchmark reference scores, and Student is the mixed-domain SFT baseline. Bold marks the best fusion result in each column.

Method	Math		Code		Science	IF		Tool use	Raw	Norm.	
	AIME24	AIME25	LCBv5	LCBv6	GPQA-D	IFEval	IFBench	BFCLv3	Avg.	Avg.	Min.
Experts	74.6	67.9	47.3	43.5	52.9	84.6	42.0	67.5	60.0	1.000	1.000
Fisher	60.4	41.0	30.6	28.8	45.2	64.0	26.8	29.8	40.8	0.677	0.442
Ada	69.0	65.8	55.9	48.5	48.2	73.9	41.4	31.3	54.3	0.928	0.463
TA	69.8	68.1	56.6	52.7	55.4	74.5	39.8	31.5	56.1	0.961	0.466
DARE	71.9	67.1	57.5	48.7	55.6	74.7	35.2	31.8	55.3	0.941	0.471
RegMean	52.5	42.1	37.1	35.7	54.0	72.3	29.0	31.8	44.3	0.746	0.471
Soup	53.1	39.8	41.2	36.1	57.2	75.0	31.8	34.1	46.0	0.778	0.505
TIES	69.0	63.5	58.4	44.5	52.3	73.7	43.2	34.8	54.9	0.940	0.515
DELLA	75.6	69.4	59.7	51.3	58.2	71.9	39.2	37.8	57.9	0.990	0.561
L&S	72.7	72.1	55.4	50.8	51.6	75.1	42.0	39.1	57.4	0.977	0.580
Student	52.8	51.5	39.2	36.2	31.2	67.4	30.8	63.1	46.5	0.773	0.590
Seq-KD	45.8	55.8	37.1	34.9	44.4	68.5	40.7	65.0	49.0	0.826	0.615
MOPD	67.9	59.8	45.9	42.6	45.5	80.3	46.1	66.9	56.9	0.955	0.859

Training-free methods fall short in retaining tool use. On 4B, BFCLv3 is the weakest normalized benchmark for seven of the eight training-free methods. TA scores 31.1 on BFCLv3, close

to the 29.7–29.8 range of experts from other domains and well below the tool use expert at 66.0 (Appendix Table 9). On 8B, BFCLv3 is the weakest normalized benchmark for all eight methods; even L&S, the best of these methods on tool use, scores only 39.1, compared with 67.5 for the corresponding expert.

MOPD achieves more balanced performance across domains. MOPD has 0.899 and 0.955 Avg. Norm. on 4B and 8B, below the best score in each pool, but the highest Min. Norm. among evaluated methods at 0.708 and 0.859. Student already has strong performance in tool use, which MOPD retains while scoring higher in reasoning and instruction following. In contrast, Seq-KD scores higher than Student on average but lower on some math or code benchmarks, and trails MOPD on all eight benchmarks in both pools.

Larger model scales do not guarantee better capability retention. In both the 4B and 8B experiments, several methods retain reasoning performance well, but tool use remains the least retained capability for most non-distillation methods. TA’s Avg. Norm. rises from 0.908 on 4B to 0.961 on 8B, but its Min. Norm. falls from 0.472 to 0.466. DELLA and L&S improve on both metrics, yet their 8B scores on tool use remain at only about 56% and 58% of the corresponding expert score. Although some methods achieve higher normalized scores on 8B, the difference in retention between reasoning and tool use remains.

4.2 CONTINUAL FUSION

Expert models often become available sequentially in practice, a setting captured by Continual Fusion (Tang et al., 2025b; Yang et al., 2025b). On Qwen3-4B, we evaluate TA, Continual Soup, and Continual DELLA, starting from the instruction-following expert and successively fusing the math, science, code, and tool use experts to examine capability acquisition and retention. Each step uses the current fused model and the new expert, without access to earlier experts’ individual weights. The common base model, calibration data from previously introduced domains, and retained statistics remain available. Table 7 reports stage-wise results, with joint fusion of all five experts as a reference for Soup and DELLA. Appendix B.10 details differences in initialization, effective coefficients, and statistical configurations between continual and joint fusion.

Table 7: Continual Fusion on Qwen3-4B, starting from the IF expert (Avg. Norm. 0.497, Min. Norm. 0.227) and adding one domain expert per stage. Avg. and Min. are expert-normalized scores across eight benchmarks.

Method	1: +Math		2: +Science		3: +Code		4: +Agent		Joint fusion	
	Avg.	Min.	Avg.	Min.	Avg.	Min.	Avg.	Min.	Avg.	Min.
TA	0.463	0.303	0.616	0.402	0.814	0.450	0.884	0.468	-	-
Continual Soup	0.624	0.451	0.668	0.451	0.744	0.451	0.583	0.352	0.663 (−12.1%)	0.512 (−31.2%)
Continual TIES	0.298	0.056	0.499	0.178	0.396	0.161	0.642	0.495	0.819 (−21.6%)	0.454 (+9.0%)
Continual DELLA	0.705	0.451	0.731	0.449	0.858	0.452	0.776	0.460	0.880 (−11.8%)	0.471 (−2.3%)

Continual Fusion still falls short of joint fusion. Under the current order and configurations, Continual Soup and DELLA finish 12.1% and 11.8% below their respective joint fusion references in normalized average performance. Adding the science expert improves math performance for both methods, and adding the code expert further improves it, showing that new experts can strengthen existing capabilities. However, adding the tool use expert yields little improvement in tool use while degrading math, code, and instruction following, as shown in Appendix Table 25. Their normalized average scores consequently decrease from 0.744 to 0.583 and from 0.858 to 0.776, respectively. DELLA also retains a minimum normalized score closer to its joint fusion reference, with a relative decline of 2.3% compared with 31.2% for Soup. Overall, improvements in individual capabilities do not translate into average performance matching joint fusion, highlighting the challenge of acquiring new capabilities while retaining existing ones.

4.3 ANALYSIS

Computational cost. Figure 2 reports runtime and peak GPU memory for the main experiments. On Qwen3-4B, TA takes 0.62 minutes and achieves higher Avg. Norm. than the slower DELLA and L&S, but retains tool use poorly. DELLA runs faster than TIES but requires more peak GPU memory, at 26.48 versus 16.70 GiB. Among the distillation methods, MOPD achieves higher Min. Norm., with training taking about 10.18 and 12.82 hours on 4B and 8B. Seq-KD takes about 16.11 and 23.99 hours, including offline teacher response generation and student training. Appendix B.5 gives the resource configurations and timing scope for each method.

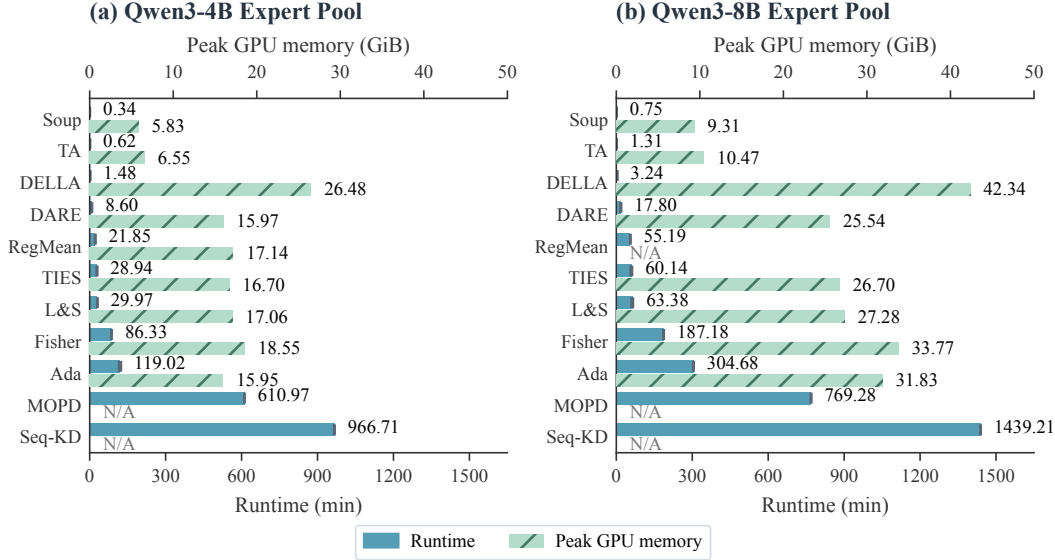


Figure 2: Runtime and peak GPU memory on Qwen3-4B and Qwen3-8B. Blue bars show mean runtime (bottom axis); hatched green bars show peak memory (top axis). Error bars indicate runtime ranges; N/A denotes unrecorded memory. Resource and timing details appear in Appendix B.5.

Examining Tool Use Degradation.

We examine whether update norm, Agent cosine similarity, and Agent-weighted sign conflict consistently track BFCLv3 screening performance (Figure 4). Within Fisher, higher BFCLv3 scores coincide with smaller update norms and higher Agent cosine similarity. Across methods, however, TIES has the largest norm but performs similarly to F1, while F2 outperforms D1 despite lower Agent cosine similarity. TIES also has lower weighted sign conflict than D1/D2 without higher BFCLv3 scores. These comparisons caution against selecting configurations based on a single update statistic. Figure 3 further shows capability tradeoffs: F2–F4 outperform F1 on BFCLv3 but score lower on AIME25 and LCBv5, with the same pattern for D3 relative to D1/D2. These changes are not uniform across reasoning domains: from F3 to F4, BFCLv3 and AIME25 both improve, whereas LCBv5 declines further. Higher tool use scores therefore do not by themselves establish better joint capability retention. Configuration selection should assess tool use and individual reasoning domains jointly rather than prioritize BFCLv3 or a single update statistic alone. Appendix B.8 provides configurations, results, and a case study.

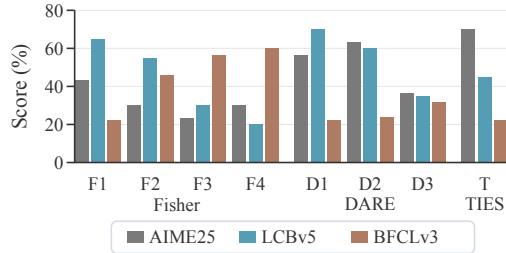


Figure 3: AIME25, LCBv5, and BFCLv3 screening scores on Qwen3-4B.

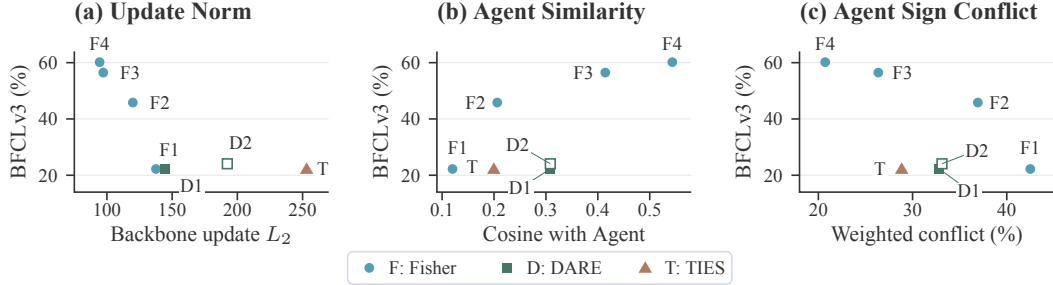


Figure 4: BFCLv3 screening performance versus update norm, Agent cosine similarity, and Agent-weighted sign conflict on Qwen3-4B. F1–F4: Fisher; D1/D2: DARE (drop rate 0.2, scales 0.3/0.4, respectively); T: TIES. See Appendix B.8 for definitions and additional configurations.

Post-Fusion Calibration. This analysis evaluates the effects of post-fusion calibration on average and worst-benchmark performance. Table 8 compares FeatCal and Surgery against TA. On 4B, Avg. Norm. falls by 0.039 and 0.023, respectively, while Min. Norm. rises by only 0.011 and 0.003. On 8B, FeatCal changes neither metric appreciably; Surgery raises Min. Norm. by 0.042 but lowers Avg. Norm. by 0.068. Gains on the weakest benchmark thus coexist with losses elsewhere, and neither calibration improves both metrics under these configurations. Full results appear in Appendix B.9.

Table 8: Effect of calibrating TA with FeatCal or Surgery. Values are changes in normalized average and worst-benchmark scores relative to TA.

Model	Calibration	Norm.	
		Avg.	Min.
Qwen3-4B	FeatCal	−0.039	+0.011
	Surgery	−0.023	+0.003
Qwen3-8B	FeatCal	−0.002	−0.001
	Surgery	−0.068	+0.042

4.4 FUTURE DIRECTIONS

Preserving reasoning and tool use in training-free fusion. Our results show that several training-free methods retain much of the math and code performance but substantially degrade tool use. Update magnitude, similarity, and sign conflict do not fully explain these changes individually. Future work should test whether preserving updates from the tool use expert in selected layers or parameter groups can reduce these losses without sacrificing reasoning. Selection could be guided by capability retention on separate validation data rather than weight statistics alone. The goal is joint retention without additional training, not higher average scores with degraded tool use.

MOPD is effective but costly. MOPD achieves a more balanced capability profile and retains tool use, but requires additional data preparation, inference, and student training compared with training-free fusion. Since the student already performs well on tool use before distillation, future work could select teachers and prompts for weaker domains while checking that this existing strength is preserved. Another direction is to combine parameter fusion with selective distillation to recover degraded capabilities using fewer teacher queries and training steps. These approaches should be evaluated by total preparation, inference, and training costs, alongside capability retention and gains over the initial student.

Continual model fusion remains challenging. All three continual methods finish below joint fusion in average performance, and adding the tool use expert yields little improvement in tool use for Continual Soup and DELLA. Future work should test whether preserving updates from the tool use expert in selected layers or parameter groups can reduce these losses without sacrificing reasoning, with global reweighting of that expert as a baseline.

5 CONCLUSION

We introduce FUSIONEVAL to address the limited coverage of long-chain reasoning and tool use in existing model fusion evaluations. FUSIONEVAL provides five-domain expert pools built from

Qwen3 and Llama models and compares training-free and training-based methods under shared data-access rules and eight public benchmarks, measuring capability retention, runtime, and peak GPU memory. Several methods retain strong reasoning performance but incur substantial losses in tool use, showing that high average scores do not establish joint retention across domains. MOPD retains tool use while achieving more balanced performance, but requires additional data preparation, inference, and training. Continual Fusion can strengthen some existing capabilities as new experts arrive, but leads to an overall loss of model capabilities relative to joint fusion in our experiments. We will release the data, expert pools, model fusion scripts, and related configurations to support comparisons under the same protocol. We hope FUSIONEVAL supports the development of parameter- and data-efficient fusion methods that retain expert capabilities more fully at lower computational cost, with attention to tool use, which is vulnerable to degradation during fusion.

AI USE STATEMENT

In this work, we primarily used generative AI tools for drafting, translation, language editing, and code optimization. AI tools also assisted experimental design and result interpretation; domain teacher models generated training responses for Seq-KD. The relevant code changes were tested, and reported results were checked against experimental records. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REFERENCES

- Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. On-policy distillation of language models: Learning from self-generated mistakes. In *International Conference on Learning Representations*, pp. 21246–21263, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/5be69a584901a26c521c2b51e40a4c20-Abstract-Conference.html.
- AIME. AIME problems and solutions, 2025. URL https://artofproblemsolving.com/wiki/index.php/AIME_Problems_and_Solutions.
- Shuo Cai, Yanggan Gu, Zihao Wang, Yuanyi Wang, Yibo Yan, Wenjun Wang, Yuhang Liu, Guanghao Zhu, Sirui Huang, Ming Li, and Hongxia Yang. From parameters to behaviors: A survey of model fusion for large language models, 2026. URL <https://arxiv.org/abs/2609.19553>.
- Pala Tej Deep, Rishabh Bhardwaj, and Soujanya Poria. DELLA-merging: Reducing interference in model merging through magnitude-based sampling, 2024. URL <https://arxiv.org/abs/2406.11617>.
- Yanggan Gu, Shuo Cai, Zihao Wang, Wenjun Wang, Yuanyi Wang, Pengkai Wang, Sirui Huang, Su Lu, Jianmin Wu, and Hongxia Yang. FeatCal: Feature calibration for post-merging models, 2026. URL <https://arxiv.org/abs/2605.13030>.
- Yifei He, Yuzheng Hu, Yong Lin, Tong Zhang, and Han Zhao. Localize-and-stitch: Efficient model merging via sparse task arithmetic. *Transactions on Machine Learning Research*, 2024. URL <https://openreview.net/forum?id=9CWU8Oi86d>.
- Yifei He, Siqi Zeng, Yuzheng Hu, Rui Yang, Tong Zhang, and Han Zhao. MergeBench: A benchmark for merging domain-specialized LLMs. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-3374. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/91f7f71cb04699f387dc863da42a1fe3-Abstract-Datasets_and_Benchmarks_Track.html.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=6t0Kwf8-jrj>.
- Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. LiveCodeBench: Holistic and contamination free

- evaluation of large language models for code. In *International Conference on Learning Representations*, pp. 58791–58831, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/94074dd5a072d28ff75a76dabed43767-Abstract-Conference.html.
- Xisen Jin, Xiang Ren, Daniel Preotiuc-Pietro, and Pengxiang Cheng. Dataless knowledge fusion by merging weights of language models. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=FCnohuR6AnM>.
- Yoon Kim and Alexander M. Rush. Sequence-level knowledge distillation. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pp. 1317–1327, 2016. doi: 10.18653/v1/D16-1139. URL <https://aclanthology.org/D16-1139/>.
- Wenhan Ma, Jianyu Wei, Liang Zhao, Hailin Zhang, Bangjun Xiao, Lei Li, Qibin Yang, Bofei Gao, Yudong Wang, Rang Li, Jinhao Dong, Zhifang Sui, and Fuli Luo. MOPD: Multi-teacher on-policy distillation for capability integration in LLM post-training, 2026. URL <https://arxiv.org/abs/2606.30406>.
- Michael S. Matena and Colin A. Raffel. Merging models with fisher-weighted averaging. In *Advances in Neural Information Processing Systems*, volume 35, pp. 17703–17716, 2022. doi: 10.52202/068431-1287. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/70c26937fbf3d4600b69a129031b66ec-Abstract-Conference.html.
- Meta AI. Llama 3.2 model card. Model card, 2024. URL https://github.com/meta-llama/llama-models/blob/main/models/llama3_2/MODEL_CARD.md.
- Shishir G. Patil, Huanzhi Mao, Fanjia Yan, Charlie Cheng-Jie Ji, Vishnu Suresh, Ion Stoica, and Joseph E. Gonzalez. The berkeley function calling leaderboard (BFCL): From tool use to agentic evaluation of large language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 48371–48392, 2025. URL <https://proceedings.mlr.press/v267/patil25a.html>.
- Valentina Pyatkin, Saumya Malik, Victoria Graf, Hamish Ivison, Shengyi Huang, Pradeep Dasigi, Nathan Lambert, and Hannaneh Hajishirzi. Generalizing verifiable instruction following. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-1645. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/46499a0622ecf568b72d17b61e45dbd5-Abstract-Datasets_and_Benchmarks_Track.html.
- David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A graduate-level google-proof Q&A benchmark. In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=Ti67584b98>.
- Wenju Sun, Qingyong Li, Wen Wang, Yang Liu, Yangli ao Geng, and Boyang Li. Towards minimizing feature drift in model merging: Layer-wise task vector fusion for adaptive knowledge integration. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-5745. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/fbb4f2269267de23e0240e1c01242981-Abstract-Conference.html.
- Derek Tam, Yash Kant, Brian Lester, Igor Gilitschenski, and Colin Raffel. Realistic evaluation of model merging for compositional generalization, 2024. URL <https://arxiv.org/abs/2409.18314>.
- Anke Tang, Li Shen, Yong Luo, Enneng Yang, Han Hu, Lefei Zhang, Bo Du, and Dacheng Tao. FusionBench: A unified library and comprehensive benchmark for deep model fusion. *Journal of Machine Learning Research*, 26(307):1–38, 2025a. URL <https://jmlr.org/papers/v26/25-1243.html>.

- Anke Tang, Enneng Yang, Li Shen, Yong Luo, Han Hu, Lefei Zhang, Bo Du, and Dacheng Tao. Merging on the fly without retraining: A sequential approach to scalable continual model merging. In *Advances in Neural Information Processing Systems*, volume 38, pp. 82081–82118, 2025b. doi: 10.52202/085713-2747. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/7611a3cb5d733e628081431445cb01fd-Abstract-Conference.html.
- Haoqing Wang, Xiang Long, Ziheng Li, Yilong Xu, Tingguang Li, and Yehui Tang. To mix or to merge: Toward multi-domain reinforcement learning for large language models. In *Conference on Language Modeling*, 2026. doi: 10.48550/arXiv.2602.12566. URL <https://arxiv.org/abs/2602.12566>.
- Ke Wang, Nikolaos Dimitriadis, Guillermo Ortiz-Jimenez, François Fleuret, and Pascal Frossard. Localizing task information for improved model merging and compression. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 50268–50287, 2024. URL <https://proceedings.mlr.press/v235/wang24k.html>.
- Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 23965–23998, 2022. URL <https://proceedings.mlr.press/v162/wortsman22a.html>.
- Prateek Yadav, Derek Tam, Leshem Choshen, Colin A. Raffel, and Mohit Bansal. TIES-merging: Resolving interference when merging models. In *Advances in Neural Information Processing Systems*, volume 36, pp. 7093–7115, 2023. doi: 10.52202/075280-0310. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/1644c9af28ab7916874f6fd6228a9bcf-Abstract-Conference.html.
- Prateek Yadav, Tu Vu, Jonathan Lai, Alexandra Chronopoulou, Manaal Faruqui, Mohit Bansal, and Tsendsuren Munkhdalai. What matters for model merging at scale? *Transactions on Machine Learning Research*, 2025. ISSN 2835-8856. URL <https://openreview.net/forum?id=9sbtmNpW>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report, 2025a. URL <https://arxiv.org/abs/2505.09388>.
- Enneng Yang, Li Shen, Zhenyi Wang, Guibing Guo, Xiaojun Chen, Xingwei Wang, and Dacheng Tao. Representation surgery for multi-task model merging. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 56332–56356, 2024a. URL [https://proceedings.mlr.press/v235/ang24t.html](https://proceedings.mlr.press/v235/yang24t.html).
- Enneng Yang, Zhenyi Wang, Li Shen, Shiwei Liu, Guibing Guo, Xingwei Wang, and Dacheng Tao. AdaMerging: Adaptive model merging for multi-task learning. In *International Conference on Learning Representations*, 2024b. URL <https://openreview.net/forum?id=nHESwXvxWK>.
- Enneng Yang, Anke Tang, Li Shen, Guibing Guo, Xingwei Wang, Xiaochun Cao, and Jie M. Zhang. Continual model merging without data: Dual projections for balancing stability and plasticity. In *Advances in Neural Information Processing Systems*, volume 38, pp. 39275–39305, 2025b. doi: 10.52202/085713-1311. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/37d9f19150fce07bcd2a81fc87d47a6-Abstract-Conference.html.
- Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. Language models are super mario: Absorbing abilities from homologous models as a free lunch. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pp. 57755–57775, 2024. URL <https://proceedings.mlr.press/v235/yu24p.html>.

Xinyu Zhao, Guoheng Sun, Ruisi Cai, Yukun Zhou, Pingzhi Li, Peihao Wang, Bowen Tan, Yexiao He, Li Chen, Yi Liang, Beidi Chen, Binhang Yuan, Hongyi Wang, Ang Li, Zhangyang Wang, and Tianlong Chen. Model-GLUE: Democratized LLM scaling for a large model zoo in the wild. In *Advances in Neural Information Processing Systems*, volume 37, pp. 13349–13371, 2024. doi: 10.52202/079017-0426. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/180a476f22a52b8ef14f42b2b927afcc-Abstract-Dataset_s_and_Benchmarks_Track.html.

Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models, 2023. URL <https://arxiv.org/abs/2311.07911>.

A EXPERT POOL RESULTS

Table 9: Cross-domain expert scores (%) on Qwen3-4B. Bold marks column maxima.

Expert	Math		Code		Science	IF		Tool use
	AIME24	AIME25	LCBv5	LCBv6	GPQA-D	IFEval	IFBench	BFCLv3
Math	73.4	67.4	21.7	22.0	39.7	48.3	21.0	29.7
Code	46.5	45.4	44.4	41.6	31.8	58.7	27.3	29.7
Science	47.5	45.2	15.6	15.8	51.9	44.6	20.8	29.8
IF	16.7	17.1	14.2	14.1	20.2	86.0	38.6	29.8
Agent	17.1	13.3	10.8	13.9	31.7	51.7	23.1	66.0

Table 10: Cross-domain expert scores (%) on Qwen3-8B. Bold marks column maxima.

Expert	Math		Code		Science	IF		Tool use
	AIME24	AIME25	LCBv5	LCBv6	GPQA-D	IFEval	IFBench	BFCLv3
Math	74.6	67.9	28.5	25.6	41.9	60.9	24.8	29.7
Code	58.8	57.9	47.3	43.5	42.9	68.6	31.7	29.3
Science	40.6	35.4	24.2	25.0	52.9	54.0	21.3	28.4
IF	29.4	24.0	22.0	24.2	40.2	84.6	42.0	29.8
Agent	22.3	20.6	14.9	15.7	46.7	62.0	23.0	67.5

A.1 LLAMA-3.2-3B EVALUATION

To examine whether the Qwen3 findings hold in another model family, we compare TA, TIES, DARE, DELLA, Consensus TA (Wang et al., 2024), and TA+FeatCal on the Llama-3.2-3B expert pool. Table 12 reports the results, with method configurations in Appendix Table 21.

Table 11: Cross-domain expert scores (%) on Llama-3.2-3B. Bold marks column maxima.

Expert	Math		Code		Science	IF		Tool use
	AIME24	AIME25	LCBv5	LCBv6	GPQA-D	IFEval	IFBench	BFCLv3
Math	32.2	31.2	2.1	2.6	11.6	22.8	13.1	21.0
Code	2.2	1.3	21.8	23.1	8.3	34.5	19.0	7.8
Science	5.0	2.1	0.9	1.9	35.0	25.3	13.2	8.8
IF	0.2	0.2	1.3	1.9	9.7	77.5	29.8	4.1
Agent	0.0	0.0	0.9	1.3	23.9	38.4	13.4	62.6

As in the Qwen3 parameter fusion experiments, tool use remains the least retained capability for most methods, with BFCLv3 giving the lowest normalized score for five of the six methods. TIES

has the highest score on tool use among the six methods but the lowest scores on both code benchmarks, again showing a trade-off between reasoning and tool use. The difference is in reasoning retention. Some fused models outperform the corresponding experts on reasoning benchmarks in the Qwen3 experiments, whereas all six methods fall below the corresponding domain experts on all eight benchmarks on Llama, with substantial losses in math, code, and science reasoning as well. DELLA, for example, exceeds the experts on all five reasoning benchmarks on Qwen3-8B but falls below them on all five on Llama. Although it still has the highest Avg. Norm. among the six methods on Llama, the score is only 0.568, with a Min. Norm. of 0.114.

Table 12: Model fusion on Llama-3.2-3B. Benchmark scores and Raw Avg. are percentages; Norm. scores are relative to the domain experts. Bold marks the best fusion result per column.

Method	Math		Code		Science	IF		Tool use	Raw	Norm.	
	AIME24	AIME25	LCBv5	LCBv6	GPQA-D	IFEval	IFBench	BFCLv3	Avg.	Avg.	Min.
Experts	32.2	31.2	21.8	23.1	35.0	77.5	29.8	62.6	39.1	1.000	1.000
DELLA	25.0	17.7	13.8	13.2	22.6	45.6	19.4	7.1	20.5	0.568	0.114
TIES	13.5	14.8	3.9	3.1	11.6	36.9	17.0	15.7	14.6	0.355	0.132
TA	20.0	15.8	10.8	10.9	16.9	50.6	20.2	8.9	19.3	0.506	0.142
Consensus TA	16.5	14.2	12.2	11.6	17.3	52.2	23.2	9.2	19.5	0.515	0.147
DARE	20.6	14.2	15.6	15.1	19.7	48.9	21.3	9.5	20.6	0.565	0.152
TA+FeatCal	21.9	22.9	8.6	9.2	16.0	46.6	23.4	14.9	20.4	0.536	0.238

B IMPLEMENTATION DETAILS

B.1 EXPERT SFT SETTINGS

Table 13: Target training settings for domain-expert SFT.

Setting	Value
Optimizer	Adam ($\beta_1 = 0.9, \beta_2 = 0.95$)
Peak learning rate (target)	5×10^{-5}
Global batch size (target)	256
Training epochs (target)	5
Learning-rate schedule	Warmup followed by cosine decay
Loss	SFT loss averaged over supervised tokens
Precision	BF16

B.2 NOTATION

Table 14: Notation used in the main text.

Symbol	Definition
M, i	Number of experts and expert index ($M = 5$).
K, b	Number of benchmarks and benchmark index ($K = 8$).
n	Generations per question.
θ_0, θ_i	Base-model and expert parameters.
θ, θ^*	Evaluated model and fused-model parameters.
τ_i	Expert task vector, $\theta_i - \theta_0$.
j, ℓ	Parameter coordinate and layer indices.
$s_b(\theta), s_b^{\text{exp}}$	Model and domain-expert scores on benchmark b .
Raw Avg.	Mean benchmark score (%).
Avg. Norm., Min. Norm.	Mean and minimum expert-normalized scores.
λ, λ_i^ℓ	Global and per-expert, per-layer fusion coefficients.
ρ, TIES_ρ	Retained fraction and TIES aggregation operator.
p, z_i	Drop probability and binary drop mask (1: dropped).
F_i, ϵ	Fisher weights and stability constant.
m_{ij}	Localization mask (1: retained).
W_i, W^*	Expert and fused linear-layer weights.
$G_i, \tilde{G}_i$	Layer-input Gram matrix and its regularized form.
$\odot$	Element-wise multiplication.

B.3 DATA DECONTAMINATION

Exact-match filtering does not detect paraphrases or contamination in pretraining data.

Table 15: Training samples removed by exact matching against each benchmark.

Domain	Math		Code		Science	IF		Tool use
Benchmark	AIME24	AIME25	LCBv5	LCBv6	GPQA-D	IFEval	IFBench	BFCLv3
Removed	0	0	0	0	0	0	0	1

B.4 EVALUATION PROTOCOLS

Quick evaluation uses one generation per question and three BFCLv3 questions from each of 17 subsets.

All candidates use the same preset prefixes of fixed dataset snapshots. Both stages use the same snapshots, so full evaluation is not an independent held-out test.

Table 16: Sample counts and decoding settings for quick evaluation.

Benchmark	Samples	Temperature	Max. output tokens
AIME24	30	0.7	32768
AIME25	30	0.7	32768
LCBv5	20	0.2	32768
IFEval	20	0.7	8192
BFCLv3	51	0.6	32768

Table 17: Decoding settings for full evaluation.

Benchmark	Temp. / top- p	Prompt
AIME24 / AIME25	0.7 / 0.8	Qwen
LCBv5 / LCBv6	0.2 / 0.95	Neutral
GPQA-D	0.7 / 0.8	Qwen
IFEval	0.7 / 0.8	Qwen
IFBench	0.0 / 1.0	Qwen
BFCLv3	0.6 / 0.95	Qwen + tools

Quick and full evaluations use seed 42 and top-k of 20.

B.5 EFFICIENCY MEASUREMENT DETAILS

Runtime and peak GPU memory are measured on H800 GPUs. Parameter-fusion timings exclude weight loading, output saving, and checks, except that the recorded 8B RegMean run includes input reading during statistic construction. For repeated runs, we report mean runtime with min–max ranges and the largest memory peak in GiB. Some timings use configurations other than the selected evaluation checkpoints.

Distillation runtime sums the recorded rollout and training stages for MOPD, and teacher generation and student training for Seq-KD, excluding identifiable initialization, I/O, and idle intervals. Concurrent generation intervals are counted once. Both methods have a planned budget of 50 updates. MOPD uses 16 GPUs; Seq-KD uses 8 GPUs for training. Runtime covers the updates completed in the recorded runs.

B.6 HYPERPARAMETER TUNING

Candidate counts exclude full evaluations and internal optimization steps; 0 denotes a fixed configuration.

Table 18 lists the search spaces and numbers of candidate validations. These follow the original methods, MergeBench (He et al., 2025), and the configurations used in our experiments. We estimate search cost as mean Algorithm runtime multiplied by the validation count. This estimate does not replace timing a complete search.

Table 18: Fusion hyperparameter search spaces and candidate counts.

Method	Candidates	Search space
Soup	0	Uniform average
Fisher Merging	6	Number of samples per example $N \in \{4, 8\}$; damping floor $\{10^{-7}, 10^{-6}\}$; target-model favoring; Fisher normalization
AdaMerging	3	Learning rate $\{3 \times 10^{-4}, 10^{-3}\}$; initialization $\{0.2, 0.3\}$; embedding coefficient $\{\text{mean}, \text{learned}\}$; 500 optimization steps
TA	10	Scaling coefficient $\lambda \in \{0.1, 0.2, \dots, 1.0\}$
LOT-Merging	5	Scaling coefficient $\lambda_{\text{LOT}} \in \{0.4, 0.6, 0.8, 1.0, 1.2, 1.4\}$; 128 calibration samples per domain
TIES	30	Retained fraction $\rho \in \{0.1, 0.2, 0.3\}$; scaling coefficient $\lambda \in \{0.1, 0.2, \dots, 1.0\}$
DARE	30	Drop rate $p \in \{0.1, 0.2, 0.3\}$; scaling coefficient $\lambda \in \{0.1, 0.2, \dots, 1.0\}$
DELLA	30	Drop rate $p \in \{0.1, 0.2, 0.3\}$; scaling $\lambda \in \{0.1, 0.2, \dots, 1.0\}$; window fixed to 0.14
L&S	0	Retained fraction fixed to $\rho = 0.40$
RegMean	6	Shrinkage coefficient $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9, 1.0\}$; 128 calibration samples per domain

B.7 SELECTED FUSION CONFIGURATIONS

Tables 19–21 give the configurations corresponding to the reported models in each expert pool. N denotes predictive samples for Fisher estimation. Data requirements are listed in Table 3. Calibration uses 128 examples per domain; distillation uses a constant learning rate.

Table 19: Selected fusion configurations for Qwen3-4B.

Method	Configuration
Soup	Uniform average (1/5 per expert)
TA	$\lambda = 0.3$
TIES	Density 0.2; scale 1.0; disjoint mean
DARE	Drop 0.2; scale 0.3; seed 42
DELLA	Drop 0.1; window 0.14; scale 1.0; seed 42
L&S	Retained fraction 0.40
Fisher	Coefficients 0.2; predictive Fisher, $N = 8$; normalized; floor 10^{-6} (all experts); full prompts
Ada	Init. 0.3; lr 3×10^{-4} ; 500 steps; embedding mean; prompt cap 4096
LOT	scale 1.0; closed-form Linear/RMSNorm
RegMean	$\alpha = 0.7$; prefix 4096; raw Gram, FP32; mean for tied embedding/head and 1D weights
Soup+Surgery	Rank 32; lr 10^{-4} ; 250 steps
Soup+FeatCal	$\lambda = 0.01$; $\rho = 1.5$; $\alpha = 0.25$
TA+Surgery	Rank 16; lr 6×10^{-5} ; 250 steps
TA+FeatCal	$\lambda = 1$; $\rho = 1.5$; $\alpha = 0.25$
Seq-KD	50 updates; batch 512; lr 3×10^{-6} ; response-token loss; prefix 4096
MOPD	30 updates; batch 512; lr 3×10^{-6} ; Adam; temperature 1.0; top- p 1.0; output cap 32768

Table 20: Selected fusion configurations for Qwen3-8B.

Method	Configuration
Soup	Uniform average (1/5 per expert)
TA	$\lambda = 0.3$
TIES	Density 0.2; scale 0.9; disjoint mean
DARE	Drop 0.1; scale 0.3; seed 42
DELLA	Drop 0.1; window 0.14; scale 1.0; seed 42
L&S	Retained fraction 0.40
Fisher	Coefficients 0.2; predictive Fisher, $N = 4$; normalized; floor 10^{-6} (first expert); prefix 4096
Ada	Init. 0.3; lr 3×10^{-4} ; 500 steps; embedding mean; prompt cap 4096
LOT	scale 1.4; closed-form Linear/RMSNorm
RegMean	$\alpha = 0.7$; prefix 4096; raw Gram, FP32; mean for embedding and 1D weights; RegMean head
Soup+Surgery	Rank 8; lr 10^{-4} ; 250 steps
Soup+FeatCal	$\lambda = 0.1$; $\rho = 1.5$; $\alpha = 0.15$
TA+Surgery	Rank 32; lr 10^{-4} ; 500 steps
TA+FeatCal	$\lambda = 0.1$; $\rho = 1.5$; $\alpha = 0.1$
Seq-KD	50 updates; batch 512; lr 3×10^{-6} ; response-token loss; prefix 4096
MOPD	40 updates; batch 512; lr 3×10^{-6} ; Adam; temperature 1.0; top- p 1.0; output cap 32768

Table 21: Selected fusion configurations for Llama-3.2-3B.

Method	Configuration
TA	$\lambda = 0.3$
TIES	Density 0.2; scale 1.0; disjoint mean
DARE	Drop 0.2; scale 0.3
DELLA	Drop 0.1; window 0.14; scale 1.0
TA+FeatCal	$\lambda = 0.1$; $\rho = 1.5$; $\alpha = 0.25$; 128/domain
Consensus TA	TALL threshold 0.2; minimum tasks 2; scale 0.3

B.8 UPDATE GEOMETRY DETAILS

Metrics and candidates. We compare Qwen3-4B backbone updates relative to the shared base, excluding embeddings and output heads. Update magnitude is the L_2 norm; Agent similarity is the cosine between fused and Agent updates. Let Δ and Δ_A denote these updates. Agent-weighted sign conflict is

$$C_A = \frac{\sum_j |\Delta_{A,j}| \mathbf{1}[\Delta_j \Delta_{A,j} < 0]}{\sum_j |\Delta_{A,j}|}. \quad (2)$$

where j indexes parameters. The zero-update fraction uses the same denominator with the condition $\Delta_j = 0$.

For the candidates in Figures 4 and 3, F1–F4 use Fisher floors $10^{-6}, 10^{-7}, 10^{-8}, 10^{-10}$ with fixed statistics (64 calibration examples per domain, $N = 8$). D1–D3 use DARE (drop, scale) values (0.2, 0.3), (0.2, 0.4), and (0.1, 0.2); T uses TIES density 0.3 and scale 1. S denotes Soup, and A denotes TA with scale 0.3. Scores use the quick-evaluation samples in Table 16; BFCLv3 screening scores use hierarchical aggregation across subsets.

Table 22: Agent-weighted sign conflict and zero-update fractions on Qwen3-4B. Candidate labels follow the configurations defined above.

Model	Conflict (%)	Zero (%)	Model	Conflict (%)	Zero (%)
F1	42.50	1.93	D1	32.85	1.89
F2	36.93	2.11	D2	33.12	1.32
F3	26.36	2.20	D3	31.88	2.86
F4	20.73	2.01	S	31.54	2.69
T	28.85	9.74	A	32.02	1.69

Fisher response examples. Table 23 compares two questions across higher-, middle-, and lower-scoring Fisher candidates (H/M/L), with full BFCLv3 scores of 57.9/45.1/29.9.

Table 23: Fisher response summaries and evaluation outcomes on two BFCLv3 questions.

Model	Response summary	Result
multiple_0 Triangle properties for sides 5, 4, 3		
H	Structured call to <code>triangle_properties_get</code> with <code>side1=5</code> , <code>side2=4</code> , <code>side3=3</code> .	Pass
M	The same structured function call and arguments as H.	Pass
L	Identifies the correct function and sides in text, but discusses XML wrappers without producing a decodable call.	Fail
multiple_100 Highest common factor of 36 and 24		
H	Structured call to <code>math_hcf</code> with <code>number1=36</code> , <code>number2=24</code> .	Pass
M	Identifies the correct function and arguments, but switches between wrapper formats; the output is parsed as text.	Fail
L	Identifies the correct function and arguments and computes 12, but uses a <code><tools></code> wrapper; the output is parsed as text.	Fail

B.9 ADDITIONAL FUSION RESULTS

Table 24: Additional fusion and calibration results on Qwen3. Scores follow Tables 5 and 6.

Method	Math		Code		Science	IF		Tool use	Raw	Norm.	
	AIME24	AIME25	LCBv5	LCBv6	GPQA-D	IFEval	IFBench	BFCLv3	Avg.	Avg.	Min.
Qwen3-4B											
LOT	39.4	35.6	23.7	26.2	42.3	65.1	26.2	30.8	36.1	0.618	0.466
Soup+FeatCal	39.4	34.8	31.7	28.2	42.3	70.8	33.4	31.2	39.0	0.678	0.473
Soup+Surgery	43.3	32.1	28.9	28.6	44.8	65.5	29.3	31.8	38.1	0.659	0.476
TA+Surgery	66.0	56.9	45.3	45.2	46.1	73.2	39.2	31.4	50.4	0.885	0.475
TA+FeatCal	68.8	67.5	46.6	39.9	38.3	66.6	39.1	31.9	49.8	0.869	0.483
Qwen3-8B											
LOT	51.3	49.0	33.9	30.5	42.4	78.7	36.8	29.8	44.0	0.735	0.441
Soup+FeatCal	58.1	51.9	49.5	41.0	54.4	77.3	41.7	30.2	50.5	0.865	0.448
Soup+Surgery	50.4	36.9	36.6	34.4	56.7	76.6	33.4	33.0	44.7	0.756	0.489
TA+FeatCal	70.4	66.7	60.2	49.6	50.0	77.5	42.2	31.4	56.0	0.959	0.465
TA+Surgery	64.4	60.0	54.7	44.5	45.8	76.3	39.7	34.3	52.5	0.893	0.508

B.10 CONTINUAL FUSION RESULTS

Stage 0 is the IF expert; Stages 1–4 add Math, Science, Code, and Agent.

Update rules. Continual Soup assigns equal weight to the current model and the incoming expert at each step. TA starts from the same IF expert and accumulates task vectors relative to the fixed base. Let $\hat{\theta}_t$ be the model after stage t and $\pi(t)$ the incoming expert:

$$\begin{aligned}
\hat{\theta}_t^{\text{Avg}} &= \frac{1}{2}\hat{\theta}_{t-1}^{\text{Avg}} + \frac{1}{2}\theta_{\pi(t)}, \\
\hat{\theta}_1^{\text{TA}} &= \theta_0 + \lambda(\theta_{\text{IF}} - \theta_0) + \lambda(\theta_{\pi(1)} - \theta_0), \\
\hat{\theta}_t^{\text{TA}} &= \hat{\theta}_{t-1}^{\text{TA}} + \lambda(\theta_{\pi(t)} - \theta_0), \quad t \geq 2, \quad \lambda = 0.3.
\end{aligned} \tag{3}$$

Table 25: Per-benchmark scores (%) at each Continual Fusion stage on Qwen3-4B.

Method	Stage	AIME24	AIME25	LCBv5	LCBv6	GPQA-D	IFEval	IFBench	BFCLv3
IF expert	0	16.7	17.1	14.2	14.1	20.2	86.0	38.6	29.8
TA	1	24.6	20.4	14.7	17.6	37.4	51.0	21.1	29.7
	2	42.5	27.1	23.7	21.4	46.0	69.5	29.1	29.7
	3	62.9	56.4	37.3	37.8	43.4	72.6	36.1	29.7
	4	65.4	65.8	45.9	42.7	45.1	71.0	38.0	30.9
Continual Soup	1	42.3	36.5	20.8	23.3	41.4	69.7	30.5	29.8
	2	50.0	41.5	22.9	25.2	49.6	66.9	28.6	29.7
	3	53.3	45.8	42.1	35.3	41.0	62.4	30.3	29.8
	4	25.8	26.3	26.7	26.5	48.5	59.4	23.2	30.0
Continual TIES	1	26.7	27.9	2.5	3.8	11.1	28.8	17.8	29.7
	2	57.9	46.7	7.9	9.2	34.1	41.8	20.2	29.7
	3	38.8	35.4	7.2	9.5	15.7	30.9	23.7	29.7
	4	41.0	33.3	27.1	27.7	42.2	53.5	23.2	50.7
Continual DELLA	1	65.2	52.1	27.2	31.7	33.1	62.7	30.3	29.8
	2	66.7	54.8	30.8	27.9	47.2	59.9	27.3	29.6
	3	67.5	64.2	49.1	40.8	45.5	61.6	33.3	29.8
	4	58.8	47.3	42.7	36.8	48.6	58.6	30.2	30.3

C DISTILLATION DATA

Prompt sources. Table 26 lists the distillation prompt pool and counts after preprocessing; experiment-specific subsets are selected separately. MOPD loads prompts with an 8,192-token limit. Agent prompts retain tool definitions and dialogue history, without executing tool calls during training. Seq-KD uses the fixed 25,600 prompts scheduled for the first 50 steps in the original MOPD data-loading and shuffle order. Both model sizes use the same prompts, with responses generated by the corresponding 4B or 8B domain experts.

Table 26: Distillation prompt sources and counts after preprocessing.

Domain	Prompts	Sources (prompts per source)
Math	18,618	Skywork/Skywork-OR1-RL-Data (12,157); nvidia/Nemotron-RL-Math-v2 (6,461)
Code	19,954	nvidia/Nemotron-RL-coding-competitive_coding (11,997); IIGroup/X-Coder-RL-40k (7,957)
Science	19,960	nvidia/Nemotron-RL-knowledge-mcqa (19,960)
IF	19,758	allenai/IF_multi_constraints_upto5 (11,861); nvidia/Nemotron-RL-instruction_following (7,897)
Agent	8,735	NVIDIA Function-Calling Pivot (7,752); nvidia/Nemotron-Cascade-2-RL-data (983)